Sliding DCT and Wiener Filter

Mathematical Models and Methods for Image Processing

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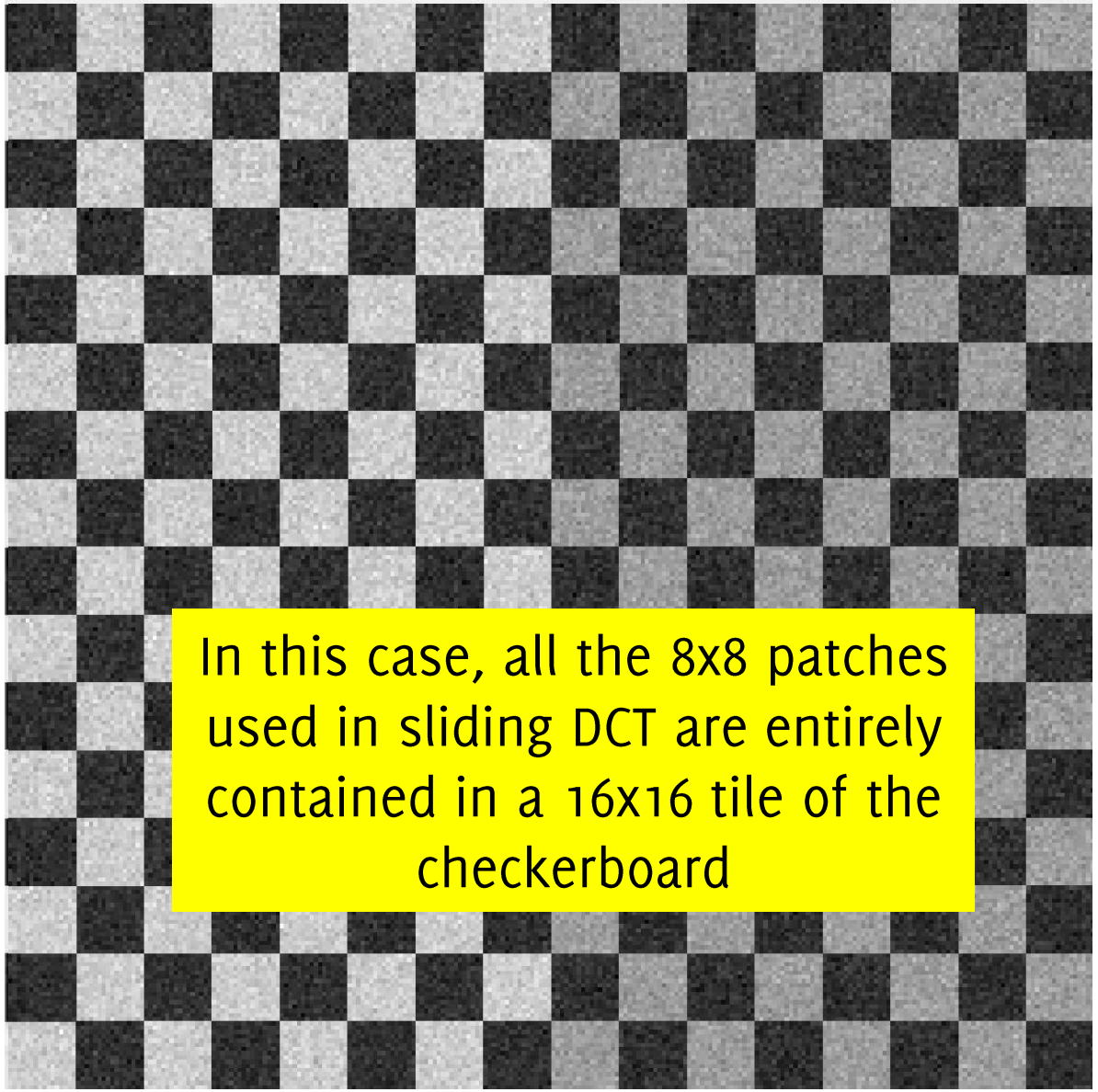
March 5th 2026

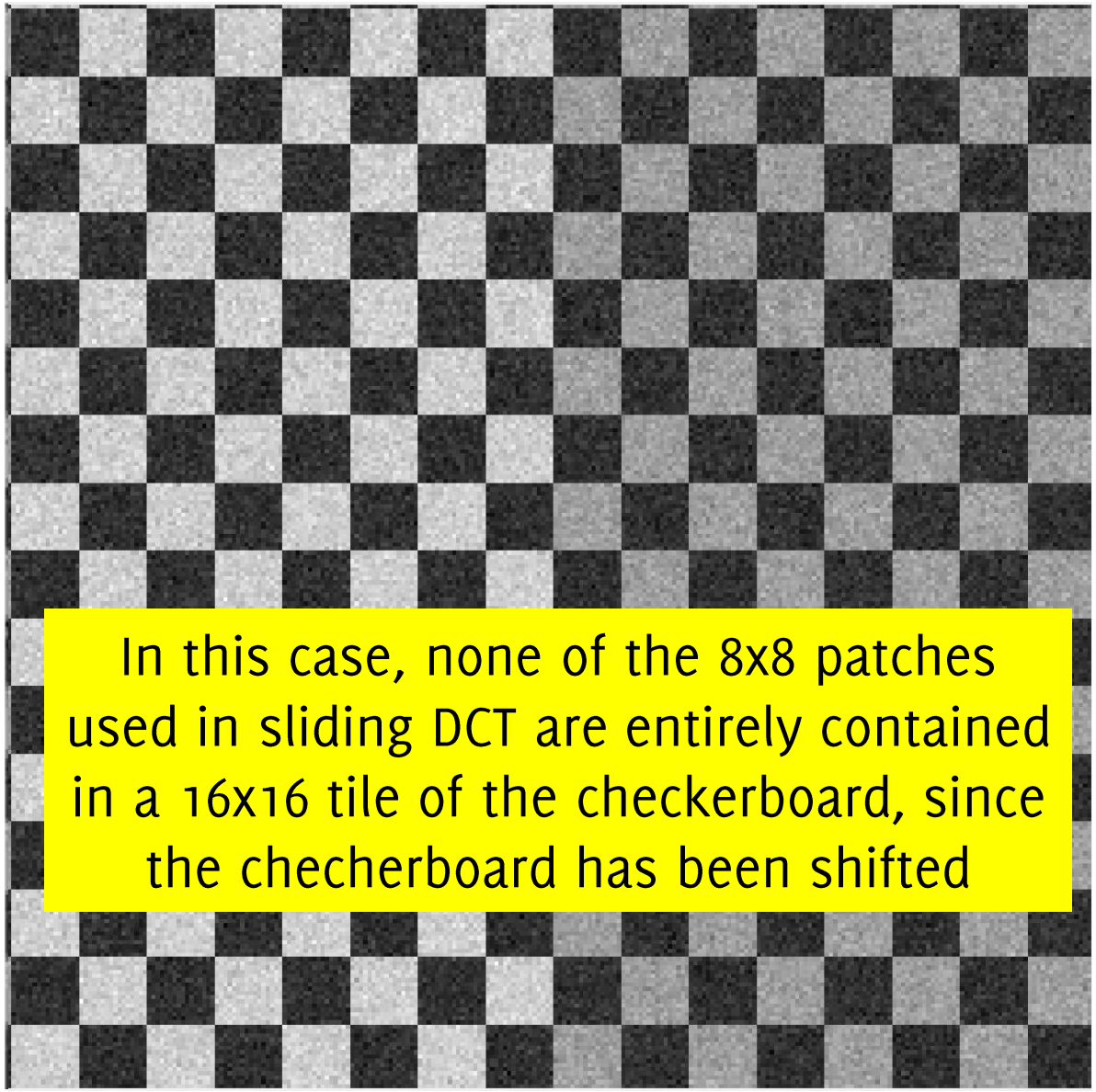
Sliding DCT

Original Checkerboard



In this case, all the 8x8 patches used in sliding DCT are entirely contained in a 16x16 tile of the checkerboard

Shift [1 row, 1 col]



In this case, none of the 8x8 patches used in sliding DCT are entirely contained in a 16x16 tile of the checkerboard, since the checherboard has been shifted

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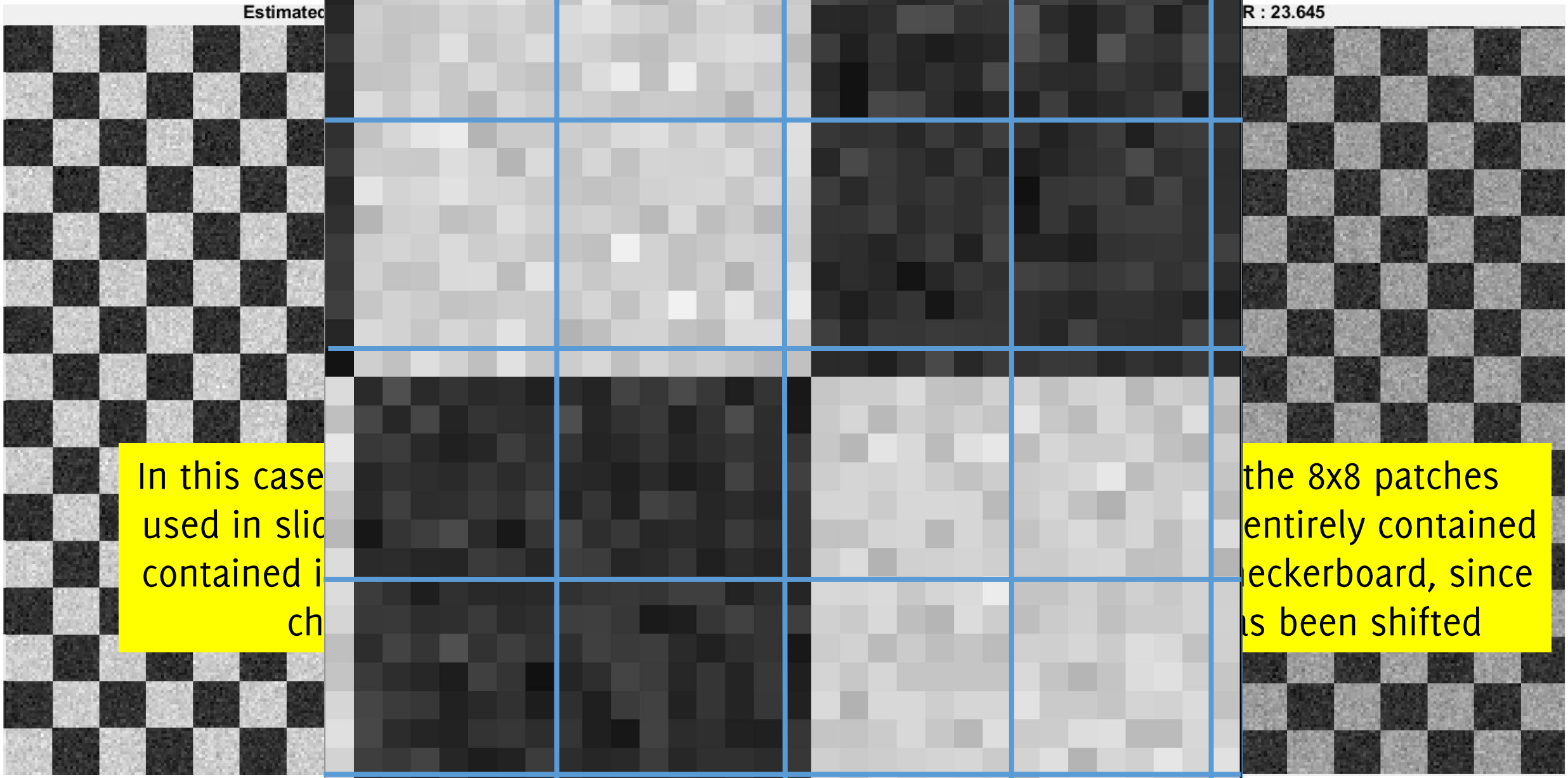
Original Chec]



In this case the 8x8 patches used in slic entirely contained contained i eckerboard, since ch s been shifted

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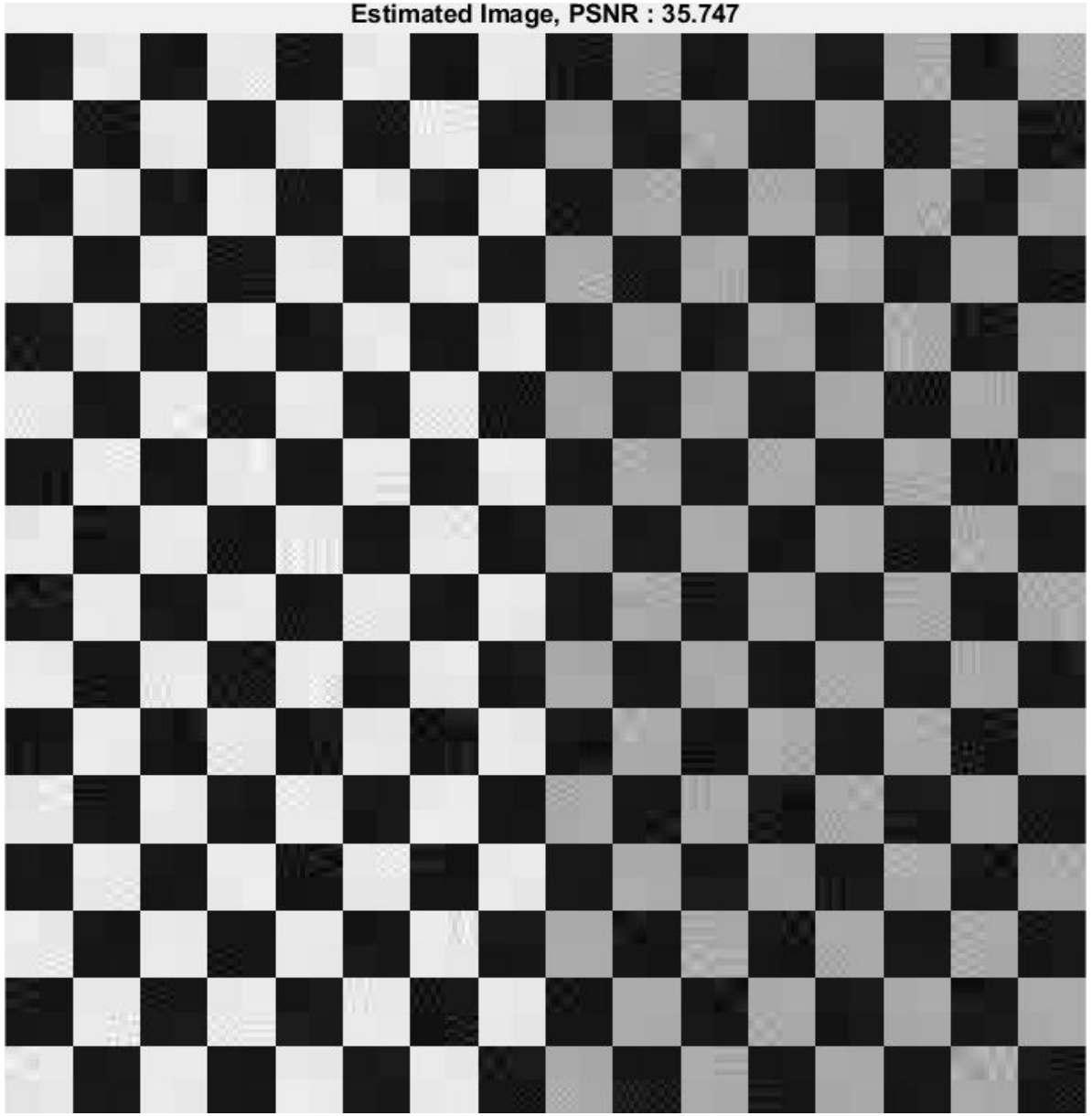
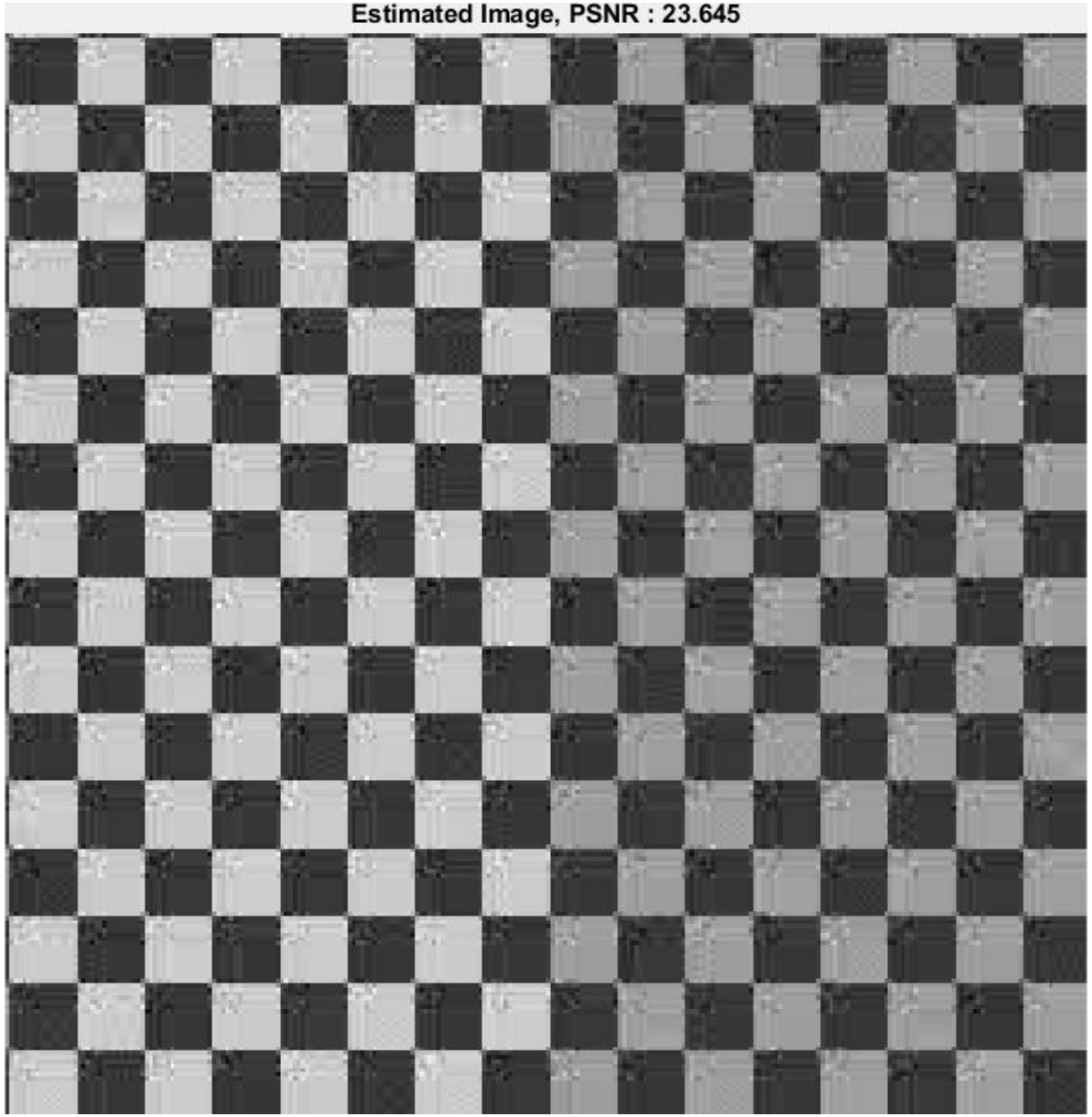
Original Checkerboard



Shift [1 row, 1 col]



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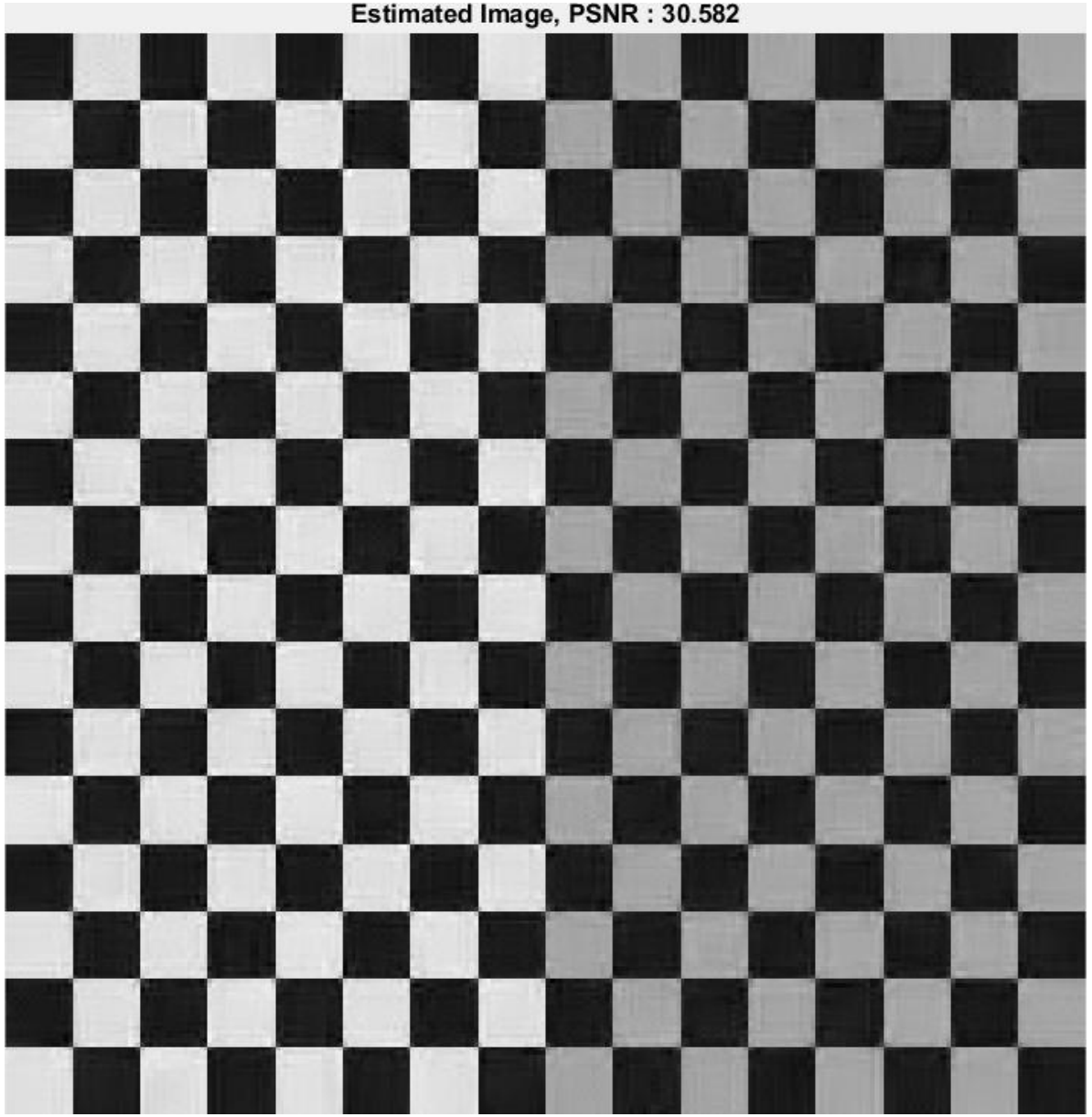
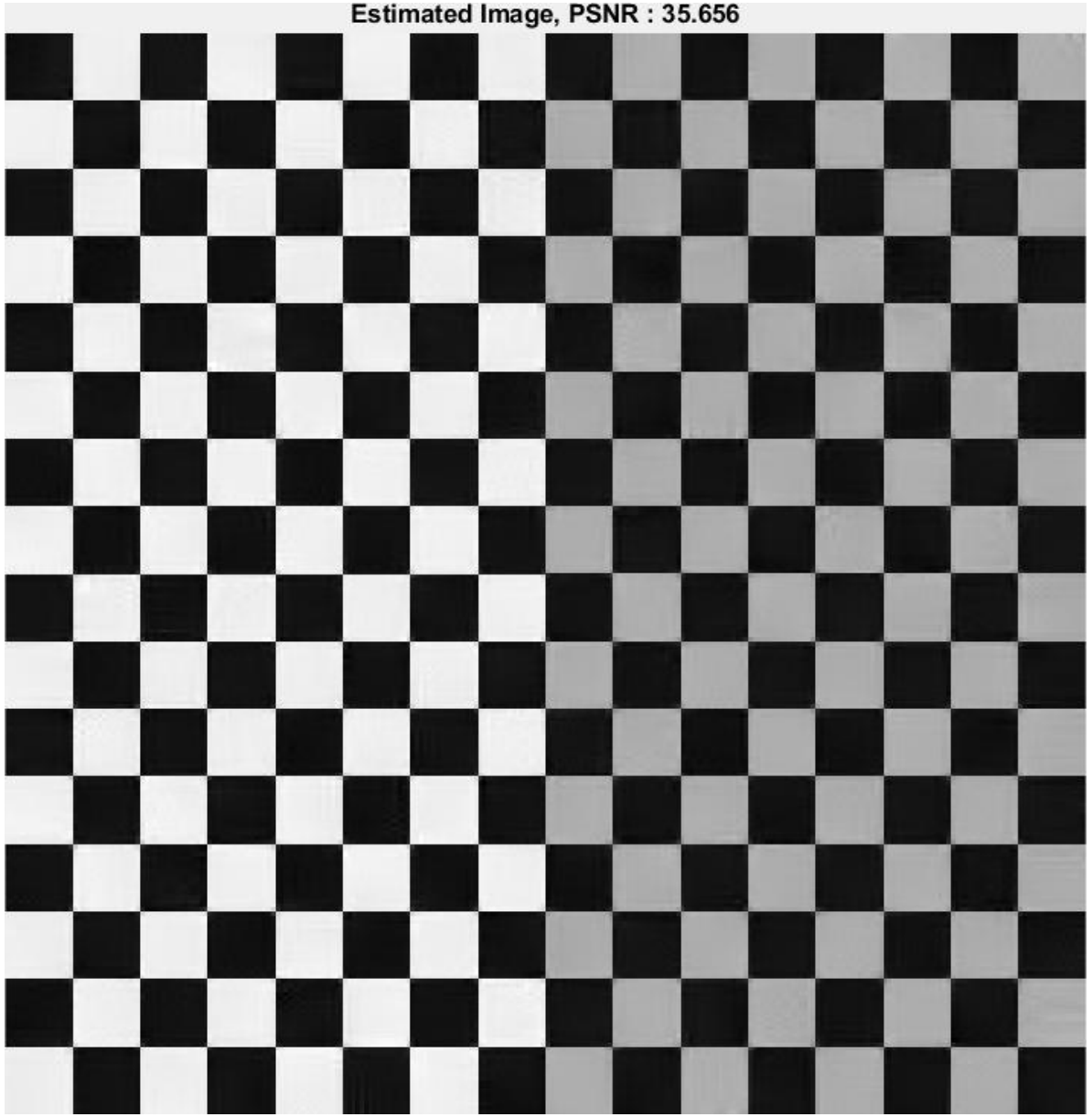
Uniform Weights



Sparsity-aware



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Wiener Filter

Empirical Wiener Filter

Let $\hat{\boldsymbol{y}}^{HT}$ be the hard threshold estimate, with DCT coefficients:

$$\hat{\boldsymbol{x}}^{HT} = D^T \hat{\boldsymbol{y}}^{HT}$$

The empirical Wiener filter attenuates the DCT coefficients as:

$$\hat{x}_i^{Wie} = \frac{\left(\hat{x}_i^{HT}\right)^2}{\left(\hat{x}_i^{HT}\right)^2 + \sigma^2} x_i$$

The empirical Wiener estimate is thus:

$$\hat{\boldsymbol{y}}^{Wie} = D\, \hat{\boldsymbol{x}}^{Wie}$$

Original image
0
50
100
150
200
250
0
50
100
150
200
250

Noisy image, PSNR = 22.09
0
50
100
150
200
250
0
50
100
150
200
250

HT Estimate, PSNR = 29.15
0
50
100
150
200
250
0
50
100
150
200
250

Wiener Estimate, PSNR = 29.41
0
50
100
150
200
250
0
50
100
150
200
250

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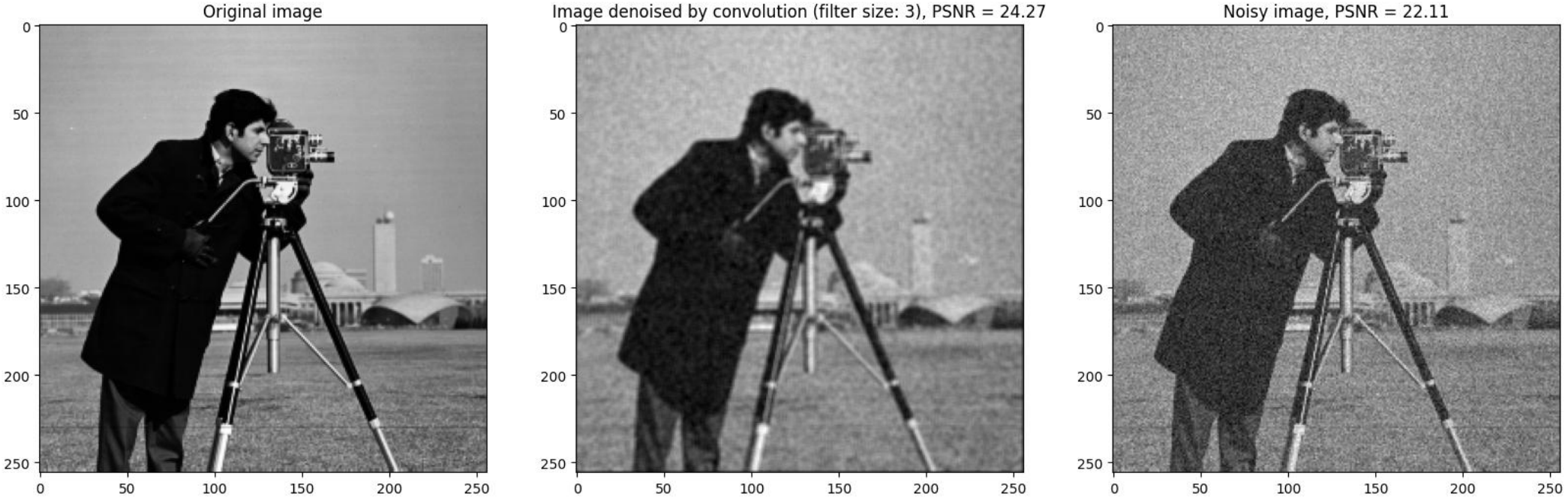
Assignments

Assignment 1– Baseline denoising

1. Synthetically generate a noisy image (AWGN model)
2. Implement the noise level estimation method
    1. compare the robust vs non-robust standard deviation estimator
3. Perform denoising by convolution
    1. what happens when you change the kernel size?



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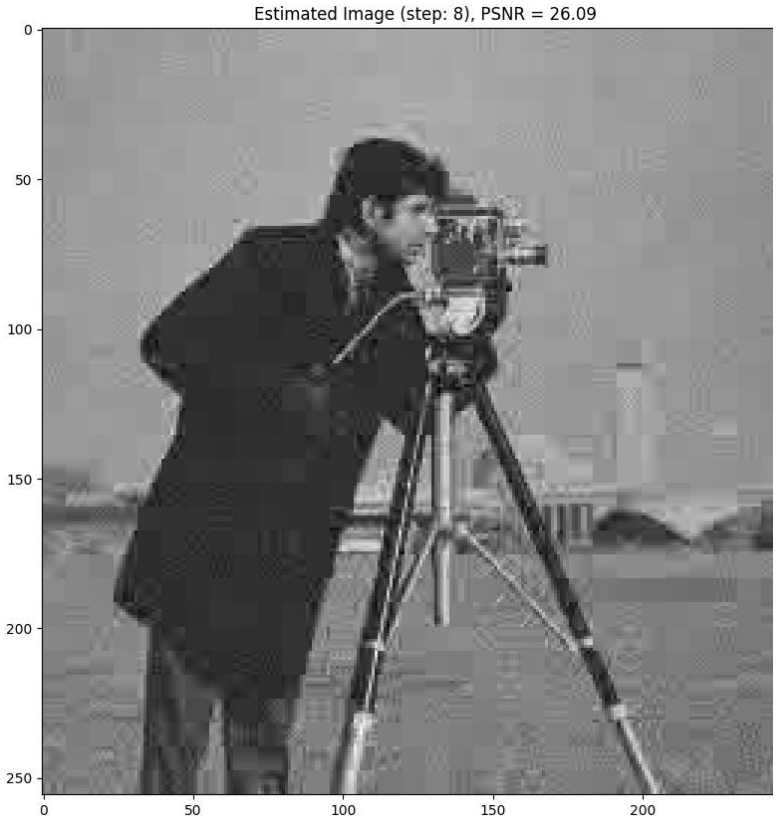
Assignment 2 – Sliding DCT denoising

1. Implement the sliding-DCT denoising using
   - no aggregation (operate on non-overlapping tiles)
   - aggregation using uniform weights
   - aggregation using weights inversely proportional to patch sparsity in DCT domain.
2. Test the three algorithms on both checkerboard and cameraman image
3. Test how much the choice of the threshold τ influences the denoising performance. Observe the resulting image when:
   - $\tau \ll 3\sigma$
   - $\tau \gg 3\sigma$

This is very important to understand how important is the choice of the threshold

DCT Denoising – Expected results



Uniform weights

Sparsity-aware weights

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Assignment 3 – Wiener filter

1. Compute a first estimate using DCT denoising with Hard Thresholding
2. Use this estimate to perform denoising via Wiener Filtering
   - Start with $STEP = \sqrt{M}$ (non overlapping patches, as in JPEG)
   - Decrease the $STEP$ and perform aggregation

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